Benthic Mapping of Coral Reef Areas at Varied Water Depths Using Integrated Active and Passive Remote Sensing Data and Novel Visual Transformer Models

Yan Zhou, Zhihua Mao, Zexi Mao, Xianliang Zhang, Longwei Zhang, and Haiqing Huang

***Abstract*— In recent years, various coral reef retrieval methods have experienced considerable progress due to a variety of observational instruments and innovative parameter calculation techniques. However, these labor-intensive methods are deficient in handling high-precision remote sensing mapping of coral reef benthic environments, facing challenges, including enhancing the robustness of scaled coral reef retrieval against varying water depths and complex water column conditions. To overcome these limitations, we propose a novel method for coral reef benthic mapping. Our method primarily consists of two central components: water depth extraction and coral reef information acquisition. Accurate bathymetry is critical for coral reef remote sensing inversion. To obtain more precise water depth, we propose the Bathymetry Transformer model. Our Bathymetry Transformer model aggregates vast amounts of active and passive remote sensing data, generates accurate bathymetry (with a 0.375-m RMSE, ranging from 0- to 12-m water depth) that eliminates the need for in situ examinations, and maintains a harmonious balance between greater bathymetry precision and finer spatial resolution. Based on this, water column corrections are then applied, and the results derived from various active and passive remote sensing processes, along with their respective band calculation outcomes, are fed into the proposed coral reef Transformer (CR Transformer) model to generate high-accuracy coral reef benthic mapping results. Copious experimental outcomes affirm that our CR Transformer surpasses current state-of-the-art (SOTA) methods in computational efficiency and results accuracy. Impressively, the CR Transformer achieves a notable mean intersection over union (mIoU) of 91.25% and an accuracy of 95.71% on the validation dataset.**

***Index Terms*— Bathymetry, coral reef, PlanetScope, remote sensing, visual transformer.**

I. Introduction

CORAL reefs, the most significant ecosystems in global tropical oceans, are often referred to as marine forests. They are edifices of limestone, erected by reef-building corals, calcifying algae, and other organisms capable of calcification, through a delicate balance of competition and cooperation among fauna and flora [1]. Despite occupying less than 0.2% of the total area of the oceans [2], coral reefs are home to a quarter of the world's marine life, with more than 40 000 marine species living within. Incidentally, these reefs also serve as a dwelling for between 6000 and 8000 fish species [3]. Globally, a staggering 1 billion people directly depend on these vibrant ecosystems for sustenance [4]. The goods and services provided by coral reefs are estimated to be worth U.S. $2.7 trillion per year, including a U.S. $36 billion tourism industry [5].

However, coral reefs are under threat from a host of factors, including global warming and ocean acidification [6], storms [7], overfishing [8], pollution [9], disease outbreaks [10], and predatory explosions [1]. Since the 1980 s, there have been three major global coral bleaching events. Notably, the El Niño event from 1997 to 1998 instigated the destruction of 16% of global coral reefs [11]. Emerging from 2011, with the progressive decline in coral cover, there has been an observable rise in algal cover. Simultaneously, coral reef communities are shifting from coral dominance to algae dominance [12], particularly in the Eastern Tropical Pacific, Red Sea, Gulf of Aden, Caribbean, Australia, and Brazil [13]. The third occurrence of bleaching, spanning 2015–2016, sets records for its enduring time, far-reaching scope, and fierce severity, wreaking unfathomable harm on coral reef ecosystems [14].

Therefore, the perpetual and extensive observation of the benthic community formation within coral reefs is instrumental to comprehend and manage the degenerative trajectory of coral reefs. Remote sensing [15], [16] emerges as a useful

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technology for portraying large-scale coral reef benthic biotypes, assessing species richness, tracing benthic organism dispersal patterns, and noting changes in benthic environments, offering high spatial–temporal resolution, cost-effectiveness, and long-term dynamic monitoring.

Nonetheless, spectral separability [15], which encompasses factors such as depth, water column murkiness, spectral distinctiveness, and varied benthic categories [17], may potentially compromise the precision of remote sensing retrieval in coral reefs [18], [19]. The inherent optical properties (IOPs) of water [20], combined with the disruption caused by atmospheric and watercolor components, might contribute to inaccuracies in quantitative inversion.

Furthermore, the previous methods of benthic segmentation utilized just a single form of data, leading to a low level and sparse quantity of feature extraction, which overlooked the potential exploitation of unique feature information available from multiple data sources for different substrate types. Moreover, conventional segmentation procedures [16], [21], [22], such as those employing machine learning, fall short in adequately learning high-level feature representations. This deficiency translates to subpar accuracy in sediment segmentation and poor generalizability of semantic segmentation results, curtailing the precision and proficiency of these models when tasked with large-scale coral reef mapping.

In conclusion, the following issues remain unaddressed in the contemporary research landscape concerning the remote sensing mechanism of coral reefs.

1) Existing inversion methods have neglected the impact of water depth on the extraction of coral reef information. Whether caused by surface-level ocean waves or internal turbulence (from the face to the inside of the water body), both of which can interfere with the information from the water column, leading to a shift in the path of radiation transmission, affecting the remote sensing observations of coral reefs at different depths. Further exploration is needed to ensure the consistency and accuracy of coral reef inversion at various depths.
2) Current inversion models are computationally intensive and possess application restrictions. On the one hand, high complexity in both computation and time makes it challenging to meet the needs of efficient observation for coral reefs. On the other hand, the precision of current algorithms leaves room for improvement, inhibited by the low spatial resolution and the misalignment of passive and active modes. This fails to provide detailed bathymetry and benthic reflectance, consequently impeding the high-accuracy retrieval of coral reef.

Deep learning models [23], [24], [25], [26], [27], in contrast to traditional methodologies [16], [22], [28], [29], [30], can effectively sift and leverage the high-level features of images, leading to more adept distinguishing of various benthic categories. When confronted with a large-scale classification problem, the semantic segmentation models have the ability to swiftly and accurately segment images at the pixel level, utilizing the characteristics of pixels with a particular radiative transfer mechanism, exhibiting notable generalization proficiencies.

Leveraging the high repetition frequency and on-orbit precision of ICESat-2, alongside the wide-ranging, time-extended remote sensing imagery, there is an assurance for high-precision bathymetric mapping, which offers fresh perspectives for coral reef substrate analysis. To put it another way, this study spotlights the method of merging these two types of data with the assistance of deep learning models to reduce the effects of water depth and inherent parameters on substrate reflectivity, eventually achieving a high degree of accuracy in the semantic segmentation of coral reef benthos.

The main contributions of the article are summarized as follows.

1) To maintain the consistency of coral reef remote sensing inversion at varying depths, we propose a novel coral reef retrieval method. This method has been tested on different reef sites and is capable of efficiently generating high-precision coral reef mapping results with low time complexity.
2) To obtain extremely precise bathymetry, we put forward the innovative Bathymetry Transformer model. It utilizes both active and passive remote sensing data, negating the need for in situ investigations and ultimately producing superior bathymetry.
3) To attain high-precision distribution results of coral reef benthos, we propose the coral reef Transformer (CR Transformer) model. It integrates the novel serial big box spatial pyramid pooling (SBBSPP) network, thereby boosting model's proficiency in carrying out exact pixel-level predictions and inferences.
4) This article evaluates the performance of the proposed models through a series of experiments, gauging the discrepancies in remote sensing image mapping retrievals that can be ascribed to factors such as bathymetric correction, water column, benthic reflectance, precision level, and associated computations. This article also includes a comparative study with related algorithms.

This article is structured as follows. Section II describes the research sites and datasets utilized in this study. Section III presents a detailed description of the proposed method. In Section IV, a set of experiments is conducted to validate and compare the performance of the proposed models with state-of-the-art (SOTA) algorithms. Finally, Section V summarizes the content of this article.

II. Study Areas and Data Description

A. Study Areas

As depicted in Fig. 1, the coral reef regions encompass the Rose Atoll of American Samoa, Xisha Islands of the South China Sea, and Guam Island. Spanning across the northern and southern hemispheres, these sites comprise a variety of reef types with each characterized by different geomorphic zones, including land, reef flat, lagoon, reef slope, reef crest, plateau, and deep water (open water), with different benthic habitats, including coral, seagrass, microalgal mats, sand, rock, and/or rubble. Including not only marginal seas

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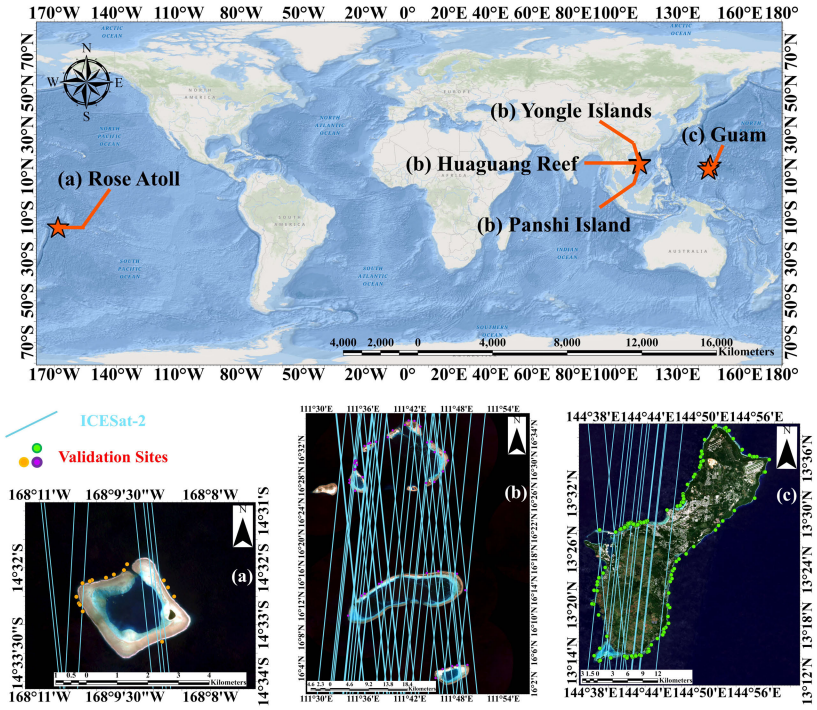


Fig. 1. Placement of coral reef benthic verification sites and the corresponding along-line photon data within the study areas.

but also the vast ocean, the variety resulted in a broad range of region signatures for testing bathymetry retrieval and coral reef semantic segmentation models.

B. Datasets and Preprocessing Workflow

To gain a comprehensive understanding of the spectral characteristics of coral reefs, ensure consistent high-level inversion across different water depths, and achieve accurate retrieval results, we conducted research combining active and passive remote sensing data.

Active ICESat-2: To obtain accurate water depth and chlorophyll-a (CHLA) data, ICESat-2 data were employed in this study. ICESat-2 [31] provides global topography measurements of lakes, reservoirs, oceans, and land surfaces. Moreover, it has the capacity to accurately detect seafloor topography up to 30 m below the surface in clear coastal regions along the satellite tracks.

1) Bathymetric Correction Along the Track (ICESAT-2): The ICESat-2 ATL03 data underwent a seven-step correction process to obtain high-precision depth: georeferencing, background noise removal (DBSCAN clustering), sea surface-seafloor segmentation (deep learning segmentation network), water column refraction correction (static coefficient: $\alpha \approx 0.7496$), water surface fluctuation correction (dynamic coefficient: ΔS), water surface slope refraction correction (dynamic coefficient: $\Delta\beta$), and tidal correction (dynamic coefficient: ΔT). The comprehensive formula for depth correction is summarized as follows (detailed in Supplementary material S1):

$$H_{\text{active}} = \Delta\beta \cdot [(\alpha \cdot I) + \Delta S] + \Delta T \tag{1}$$

where $\mathcal{I}$ is the preliminary water depth data obtained after sea surface-seafloor segmentation and H_{active} is the final water depth result after correction.

2) CHLA Retrieval Along the Track (ICESAT-2): All study areas are Case-I waters, where phytoplankton primarily determines the optical properties of the water column. The diffuse attenuation coefficient can be effectively estimated [32] based on the relationship between the subsurface signal and the active depth H_{active}

$$S(H_{\text{active}}) = C\beta'^{(\pi, H_{\text{active}})} \cdot e^{-\alpha H_{\text{active}}} \tag{2}$$

where $S(H_{\text{active}})$ denotes the ICESat-2 lidar signal at depth H_{active}, C is a calibration constant, and $\beta'^{(\pi, H_{\text{active}})}$ is the volume backscattering coefficient. The effective attenuation coefficient, α, equals the diffuse attenuation coefficient $K_d(532)$ of ICESat-2, which can be determined [33], [34] through the exponential decay relationships between the ICESat-2 lidar signal and the active depth H_{active}.

Subsequently, the CHLA concentrations (Chla) can be obtained [35] from $K_d^{\text{ICESat}}(\lambda)$, as estimated by ICESat-2

$$K_d^{\text{ICESat}}(\lambda) = K_d^{\text{Water}}(\lambda) + \chi(\lambda) \cdot \text{Chla}^{\varrho(\lambda)} \tag{3}$$

where $K_d^{\text{Water}}(\lambda)$ stands for the diffuse attenuation coefficient of pure water and $\chi(\lambda)$ together with $\varrho(\lambda)$ is the empirical parameter.

Passive Sentinel-2: Sentinel-2 is a part of the Copernicus program and aims to systematically capture high-resolution optical images (10–60 m) of land and coastal waters. The Sentinel-2 L1C data within the study areas were screened using Google Earth Engine (GEE). The ACOLITE module [36] is employed to carry out atmospheric correction.

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TABLE I
DETAILED INFORMATION OF SENTINEL-2 AND PLANETSCOPE REMOTE SENSING DATA

Sentinel-2			PlanetScope (Mosaic processed)		
Description	Pixel Size (m)	Wavelength (nm)	Description	Pixel Size (m)	Wavelength (nm)
Aerosols	60	443.9 (S2A) / 442.3 (S2B)	-	-	-
Blue	10	496.6 (S2A) / 492.1 (S2B)	Blue	4.773	490
Green	10	560 (S2A) / 559 (S2B)	Green	4.773	565
Red	10	664.5 (S2A) / 665 (S2B)	Red	4.773	665
NIR	10	835.1 (S2A) / 833 (S2B)	Near-Infrared	4.773	865

Passive PlanetScope: To assess the robustness and generalizability of the proposed algorithm, PlanetScope [37] data (accessed on May 1, 2024, https://www.planet.com/) were utilized. We carefully chose and processed PlanetScope scenes, ensuring that they were free from cloud and ocean wave interference. To facilitate the investigation of coral reef habitats, the best scene-on-top (BOT) techniques were utilized to create a series of mosaics of PlanetScope imagery. More detailed information is provided in Table I. It is worth noting that Sentinel-2 imagery was only used for Guam Island, while the PlanetScope data were employed for the rest of the study areas.

3) Bathymetric Correction Over a Region (Sentinel-2 and PlanetScope): After atmospheric correction, the remote sensing reflectance $R_{rs}(\lambda)$ can be computed from the surface reflectance $\rho(\lambda)$

$$R_{rs}(\lambda) = \frac{\rho(\lambda)}{\pi}. \tag{4}$$

In the study area, sun glint was corrected through the following equation:

$$R_{rs,0+} = R_{rs}(\lambda) - R_{rs}(\text{NIR}) \tag{5}$$

where $R_{rs,0+}$ represents the remote sensing reflectance above the water surface in the blue, green, and red bands, while $R_{rs}(\text{NIR})$ is the remote sensing reflectance in the near-infrared band.

Subsequently, the mask was established using the NIR band as it provides the highest contrast in reflectance for distinguishing land from water. Through the normalized difference water index (NDWI) [38], the ocean area is delineated from the corrected satellite images

$$\text{NDWI} = \frac{R_{rs}(\text{Green}) - R_{rs}(\text{NIR})}{R_{rs}(\text{Green}) - R_{rs}(\text{NIR})} \tag{6}$$

where $R_{rs}(\text{Green})$ denotes the remote sensing reflectance of the green band.

Then, the beneath-surface remote sensing reflectance, denoted as $r_{rs}(\lambda)$, is computed from $R_{rs,0+}$ with the aim to eliminate the influence of the air-water interface

$$r_{rs} = \frac{R_{rs}(\lambda)}{0.52 + 1.7 \cdot R_{rs}(\lambda)}. \tag{7}$$

Based on the multiband spectral of passive remote sensing images, the band-ratio algorithm [39] is employed to extrapolate the depth of the water H_{passive}

$$H_{\text{passive}} = M_1 \times \frac{\ln(r_{rs} \times \lambda_1)}{\ln(r_{rs} \times \lambda_2)} - M_0 \tag{8}$$

where M_0, M_1, λ_1, and λ_2 are all adjustable parameters. Depending on the attenuation conditions of the water column, the adjustable constants within the study area can be initially calibrated. λ_1 and λ_2 are set as constant 1000 (typically ranging from 500 to 1500) to carry out preliminary passive inversion of water depth. M_0 and M_1 can take the following values:

$$M_0 = 52.073 \times e^{(0.957 \times \text{Chla})} \tag{9}$$

$$M_1 = 50.156 \times e^{(0.957 \times \text{Chla})}. \tag{10}$$

4) Benthic Substrate Reflectance Estimation Over a Region (Sentinel-2 and PlanetScope): In the optically shallow water area, the remote sensing reflectance comes from a collective contribution of the water body and the bottom sediments. Hence, the below-surface remote sensing reflectance, denoted as r_{rs}, can be modeled [40] as follows:

$$r_{rs} = r^c + r_{rs}^b = r_{rs}^{dp}\left(1 - e^{-D_c(a+b_b)H}\right) + Be^{-D_b(a+b_b)H} \tag{11}$$

where r^c represents the contribution from the water column, r_{rs}^b represents the contribution from the underwater bottom sediments, H is the estimated water depth, and B is the bottom reflectance to be derived. Total absorption is denoted by a, and all backward scattering is represented by b_b. The term $D(a + b_b)H$ accounts for light attenuation caused by absorption and backscattering in the water column. This includes the light fractions from the water column (D_c) or those originating from the bottom (D_b). Empirical dynamics D_c and D_b correlate with an elongated underwater path of photons caused by scattering, calculated as follows:

$$D_c = 1.03 \times \left(1 + 2.4 \times \frac{b_b}{a + b_b}\right)^{0.5} \tag{12}$$

$$D_b = 1.05 \times \left(1 + 5.5 \times \frac{b_b}{a + b_b}\right)^{0.5}. \tag{13}$$

In situations where the water depth is infinite, the below-surface remote sensing reflectance labeled as r_{rs}^{dp} is constructed as

$$r_{rs}^{dp} = \left(0.089 + 0.125 \times \frac{b_b}{a + b_b}\right)\frac{b_b}{a + b_b}. \tag{14}$$

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Subsequently, the IOPs of water are modeled based on its various components [20]. The IOPs of water [41] involve contributions from pure water body [$a_w(\lambda)$ and $b_{\mathrm{bw}}(\lambda)$] and colored dissolved organic matter (CDOM) ($a_{\mathrm{cdom}}(\lambda)$), along with particles [$a_p(\lambda)$ and $b_{\mathrm{bp}}(\lambda)$], as demonstrated in the following:

$$a(\lambda) = a_w(\lambda) + a_p(\lambda) + a_{\mathrm{cdom}}(\lambda) \tag{15}$$

$$b_b(\lambda) = b_{\mathrm{bw}}(\lambda) + b_{\mathrm{bp}}(\lambda). \tag{16}$$

Combining the formulas above, the bottom reflectance B can be derived

$$B = \left(r_{\mathrm{rs}} - r_{\mathrm{rs}}^{\mathrm{dp}}\right) \times e^{D_b(a+b_b)H} + r_{\mathrm{rs}}^{\mathrm{dp}} \times e^{(D_b - D_c)(a+b_b)H}. \tag{17}$$

The quasi-analytical algorithm-RGB (QAA-RGB) [42] was established to compute the absorption coefficient, scattering coefficient, and attenuation coefficient of seawater. These coefficients solely associate with the composition of the water body and remain unaffected by variations in light conditions.

C. Coral Reef Validation Datasets

The coral reef validation datasets used in this study include in situ water depth data and a remote sensing semantic segmentation dataset. Due to the large study area, a stratified random sampling design was employed to survey the coral reef ecosystems across these regions. The stratification scheme included factors such as reef zone benthos, and depth. Specifically, as shown in Fig. 1(a)–(c), depth measurements and corresponding continuous benthic photography were conducted at various depths near the validation sites in each study area.

To conduct stratified random sampling, depth classification was established as shallow, mid, and deep, based on the midpoint between the minimum and maximum depths of the transects at each site. The in situ depth data from the stratified random sampling were used for the Bathymetry Transformer.

As depicted in Fig. 2, these benthic images, which included depth information (depth tags), were analyzed and combined with remote sensing images (PlanetScope/Sentinel-2) to generate a remote sensing semantic segmentation dataset for coral reefs at various water depths. The remote sensing semantic segmentation dataset includes ground-truth labels and original satellite images for training and validating the CR Transformer. The benthic categories of coral reef habitats are classified into five types: coral, algae, rubble/rock, sand, and unknown.

III. Proposed Method

This section provides a systematic introduction to the proposed Bathymetry Transformer and CR Transformer semantic segmentation models, encompassing vital aspects such as the creation of ultraprecise water depth outputs and the dissection of coral reef substrate types.

A. Overview

To gain a better understanding of the benthic distribution of coral reefs, various types of active and passive data sources, including ICESat-2 ATLAS, PlanetScope, Sentinel-2, and sample data, are integrated for coral reef substrate segmentation mapping. The technological pathway of this study is depicted in Fig. 3.

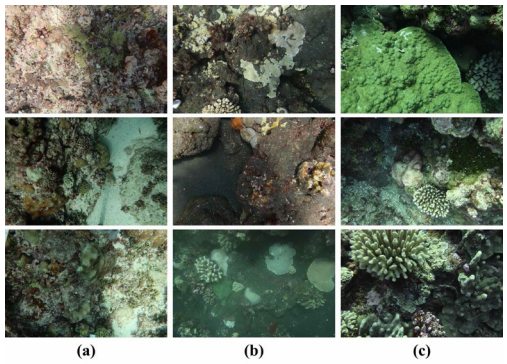


Fig. 2. Schematic of coral reef validation datasets. Classification of depth (shallow, mid, and deep) based on midpoint between minimum and maximum depths of transect(s) of given sites. (a) Shallow. (b) Mid. (c) Deep.

In a nutshell, we conducted a seven-step calibration on ICESat-2 ATLAS to obtain active water depth (H_{active}) and Chla results. In addition, preprocessing and band-ratio water depth inversion were employed on PlanetScope and Sentinel-2 data to retrieve passive water depth (H_{passive}) and benthic reflectance results. A series of active and passive water depth data is inputted into the Bathymetry Transformer for training and validation, ultimately generating high-precision water depth results (H_{final}). Subsequently, we created a semantic segmentation dataset for coral reef benthic types using sampled data collected from coral reef areas. Various active and passive remote sensing preprocessing results, along with their respective band calculation outcomes, are inputted into the CR Transformer model to generate high-accuracy semantic segmentation results, culminating in the creation of a detailed map depicting the benthic composition in the study areas.

B. Bathymetry Transformer Model for Bathymetry Retrieval

Drawing from the successful applications of unique Transformer models in spatiotemporal prediction scenarios [26], [27], [43], and acknowledging the distinctive coupling inherent in bathymetry-related active–passive remote sensing, a computationally efficient spatiotemporal attention-based neural network model, dubbed the Bathymetry Transformer, has been designed to primary bathymetry retrieval. The water depth H_{passive}, obtained from passive remote sensing imaging, will be combined with water depth H_{active}, procured from active lidar, to facilitate the generation of highly precise depth data.

In line with most high-performing transduction models, the Bathymetry Transformer model is also rooted in an encoder–decoder framework, co-involving modules such as two data preprocessing ones, key encoder and decoder components, and an output layer, as shown in Fig. 4.

The Bathymetry Transformer model utilizes 28 months (M_{in}, from September 2020 to December 2022) of gridded multivariate bathymetry H_{passive} data as input predictors (referred to here as $\mathfrak{V}^{\mathrm{in}}_{M_{\mathrm{in}} \times \mathfrak{K} \times \mathfrak{P}_{\mathrm{lat}} \times \mathfrak{P}_{\mathrm{lon}}}$), consisting of passive remote sensing retrieval bathymetry fields (hence, the channel

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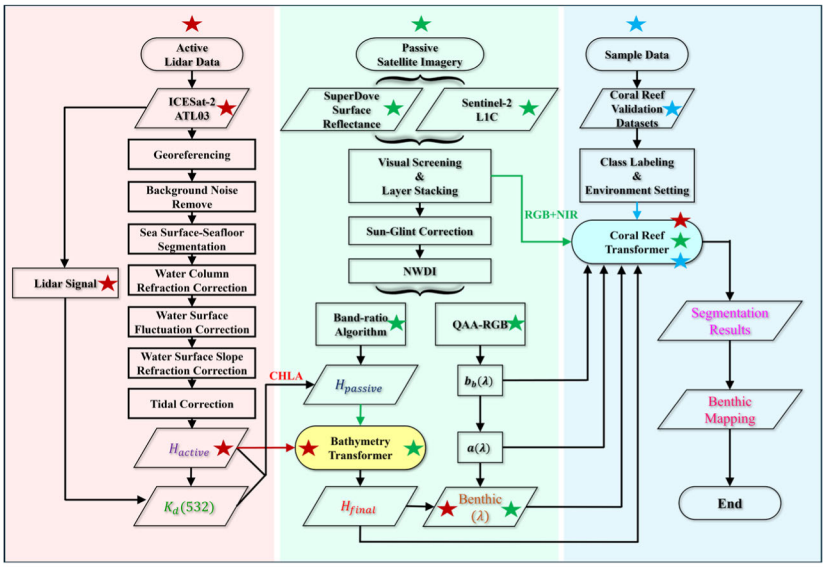


Fig. 3. Schematic of an integrated coral reef mapping method using active and passive remote sensing data with novel visual transformers.

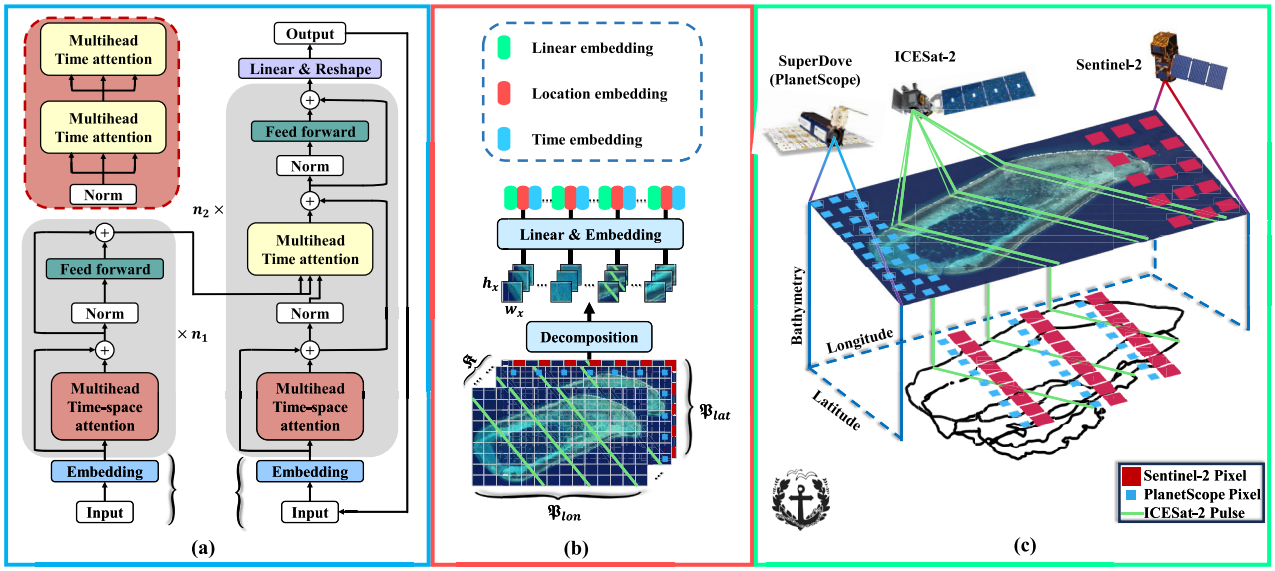


Fig. 4. Proposed Bathymetry Transformer model. (a) Structure of Bathymetry Transformer model. (b) Detailed preprocessing module and patch embedding process of Bathymetry Transformer. (c) Schematic of active and passive remote sensing fusion deployed in Bathymetry Transformer.

axis is set to $\mathfrak{K} = 2$, with one for PlanetScope and another for Sentinel-2) with a dimension of $M_{\text{in}} \times \mathfrak{K} \times \mathfrak{P}_{\text{lat}} \times \mathfrak{P}_{\text{lon}}$, in which $\mathfrak{P}_{\text{lat}} \times \mathfrak{P}_{\text{lon}}$ represents the total number of space grid points in the original fields. The fields for the following 15 months with dimensions of $M_{\text{out}} \times \mathfrak{K} \times \mathfrak{P}_{\text{lat}} \times \mathfrak{P}_{\text{lon}}$ are the targeted predictands (denoted as $\mathfrak{V}^{\text{target}}_{M_{\text{out}} \times \mathfrak{K} \times \mathfrak{P}_{\text{lat}} \times \mathfrak{P}_{\text{lon}}}$).

As illustrated in Fig. 4, prior to the encoder's engagement, the initial step involves the preprocessing module accepting inputs, specifically the predictors $\mathfrak{V}^{\text{in}}_{M_{\text{in}} \times \mathfrak{K} \times \mathfrak{P}_{\text{lat}} \times \mathfrak{P}_{\text{lon}}}$. Subsequently, predictors from each month are fragmented into a sequence of fixed-size patches without overlap, with dimensions of $\mathfrak{K} \times h_x \times w_x$ across the channel dimension. These patches are then correspondingly mapped to a certain dimensional symbolic representation, carrying embedded spatiotemporal information [refer to Fig. 4(b)]. Following this, the symbolic representation gets ingested into the ensuing encoder. This encoder has a structure with $n_1 = 4$ identical encoding blocks, where each block comprises a multiheaded spatiotemporal attention layer (in this case, the head count is decided as 4; refer to Fig. 4(a), red block area), as well as a fully connected network. The encoder module condenses the symbolic depiction into a feature map matrix, which encapsulates multivariate data for a span of 28 sequential months. Then, the feature map from the encoder is transferred to the decoder and scrutinized across the ensuing $n_2 = 4$ decoding blocks dedicated to predictions, where each block incorporates three sublayers. Finally, all predictions are associated with the bathymetry fields $\mathfrak{V}^{\text{out}}_{M_{\text{out}} \times \mathfrak{K} \times \mathfrak{P}_{\text{lat}} \times \mathfrak{P}_{\text{lon}}}$ (i.e., H_{final}), maintaining the same spatial resolutions as predictors within the output layer.

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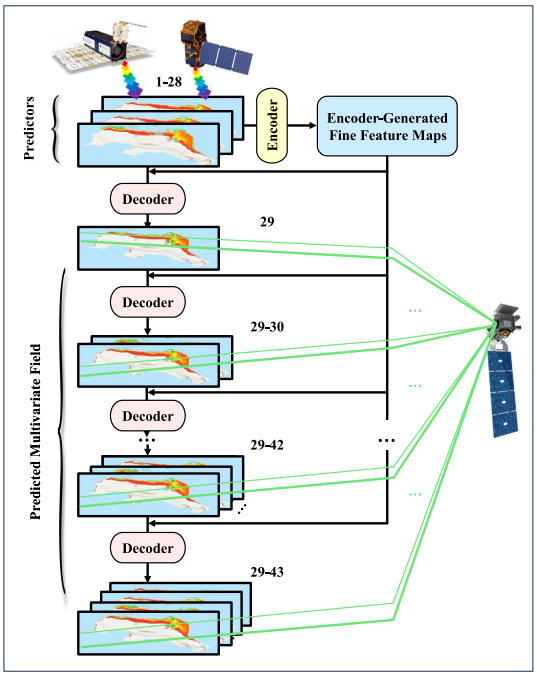


Fig. 5. Graphical representation of the Bathymetry Transformer model executing strategy.

C. Bathymetry Transformer Model Training Strategy

The Bathymetry Transformer model, equipped with a self-attention mechanism, utilizes blocks of variables (with a batch size of 8) covering a period of 28 consecutive months as primary predictors. Following that, it harnesses the forthcoming 15 months' output fields as the targeted predictands, priming the model via a dynamic rolling prediction strategy (refer to Fig. 5). Simply put, during each prediction phase, the model progressively generates the forecast for the next month by incorporating its earlier predictions, mimicking the approach used in dynamic predictive models. Simultaneously, the high-accuracy bathymetric readings obtained from ICESat-2, precisely following its orbit, function as a validation dataset. This validation dataset evaluates the recurring prediction results with the aim of formulating a high-resolution bathymetric map of the coral reef zones.

To quantify the multivariate fields crisply and boost the predictive precision in bathymetry, we architect a unified loss function. This function binds the RMSE of multivariate fields and the water depth H_{active} retrieved from ICESat-2, functioning as a scale to gauge the divergence between the predictions and the set target predictand

$$\text{Loss} = \frac{1}{M_{\text{out}}}\sum_{i=1}^{M_{\text{out}}}\left[\sqrt{\frac{1}{\mathfrak{P}_{\text{lat}}\times\mathfrak{P}_{\text{lon}}\times\mathfrak{K}}\sum_{j=1}^{\mathfrak{P}_{\text{lat}}}\sum_{k=1}^{\mathfrak{P}_{\text{lon}}}\sum_{t=1}^{\mathfrak{K}}\left(\mathfrak{V}^{\text{out}}_{i.j.k.t}-\mathfrak{V}^{\text{target}}_{i,j,k,t}\right)^2}+\sqrt{\left(\mathfrak{A}^{\text{out}}_{\text{along}_{\text{orbit}}}-\mathfrak{A}^{\text{target}}_{\text{along}_{\text{orbit}}}\right)^2}\right] \tag{18}$$

where $\mathfrak{V}^{\text{target}}_{i,j,k,t}$ represents the field of target predictand, while $\mathfrak{A}^{\text{out}}_{\text{along_orbit}}$ and $\mathfrak{A}^{\text{target}}_{\text{along_orbit}}$ symbolize the calculated output and targeted area bathymetry indices, respectively, the latter being derived from the along-orbit data of ICESat-2. To optimize the Bathymetry Transformer model, we proceed with an Adam optimization algorithm subjected to a learning rate warm-up maneuver, starting with an initial learning rate of 2×10^{-3}.

D. SBBSPP Network

For holding onto the multiscale representations of the multispectral data of the coral reef benthos, the spatial pyramid pooling (SPP) network [44] was adopted in tandem with Big Box attention [Fig. 6(b)] in a sequential style, consequently leading to the creation of an innovative SPP module termed SBBSPP network, as depicted in Fig. 6(c).

SBBSPP is composed of five branches operating both parallelly and serially, encompassing one shortcut connection, three Big Box attentions with big index B values of (2, 4, 8), and an image pooling branch. The branches of the Big Box attention furnish three parallelly and serially linked hierarchies of receptive fields for the local window, resulting in improved efficiency. In coherence with earlier studies on window attention methodologies, the patch size for the local window is preset at 8. As a result, the sizes for the receptive fields provided come in (16, 32, 64).

Compared to the conventional local window attention [45] [Fig. 6(a)], the Big Box attention employs adjustable boxes for attention computation. To address the computational perturbations resulting from the introduction of big index values B, Big Box attention applies a two-step strategy. It first compresses larger blocks and then expands smaller blocks [Fig. 6(b)], all while processing feature channels.

To be more precise, for an input feature $\mathfrak{F}^{\Delta} \in \mathbb{R}^{H\times W\times \mathfrak{C}}$, with height H, width W, $\mathfrak{C}$ channels, and stride $\mathfrak{S}$, we ascribe an intermediate convolutional layer $\mathcal{L}$, applying $\mathfrak{B}\times\mathfrak{B}$ kernel sizes, which subsequently results in an output feature map $\mathfrak{F}^{\Delta\Delta} \in \mathbb{R}^{H\times W\times(\mathfrak{C}/\mathfrak{B}^2)}$. The computational cost of $\mathcal{L}$ can be determined using the following calculation:

$$\text{Cost}(\mathcal{L}) = HW(\mathfrak{B}\times\mathfrak{B})\left(\mathfrak{C}\times\frac{\mathfrak{C}}{\mathfrak{B}^2}\right) = HW\mathfrak{C}^2. \tag{19}$$

Clearly, the computational complexity of $\mathcal{L}$ is equivalent to that of the linear mapping layer. With the big box having a shape of $(\mathfrak{S}\mathfrak{B}, \mathfrak{S}\mathfrak{B})$ and the number of patches being $((H/\mathfrak{S}), (W/\mathfrak{S}))$, the spatial memory utilized for context patches $(\propto\mathbb{R}^{((HW/\mathfrak{S}^2))(\mathfrak{S}\mathfrak{B}\times\mathfrak{S}\mathfrak{B})(\mathfrak{C}/\mathfrak{B}^2)})$ is equivalent to the spatial memory used for the query patches $(\propto\mathbb{R}^{((HW/\mathfrak{S}^2))(\mathfrak{S}\times\mathfrak{S})(\mathfrak{C})})$.

To ensure alignment between the shape of the context patch and the query patch, the output of $\mathcal{L}$ can be transformed by applying a convolutional layer $\mathcal{S}$ constructed with $\mathfrak{B}\times\mathfrak{B}$ kernel sizes. The updated context patches, after being processed by layer $\mathcal{S}$, do not incur any additional memory usage and maintain the same shape as the query patches. To be more specific, the computational costs of rearranging the context patches using $\mathcal{S}$ can be calculated as follows:

$$\text{Cost}(\mathcal{S}) = \left(\frac{HW}{\mathfrak{S}^2}\right)(\mathfrak{S}^2)(\mathfrak{B}\times\mathfrak{B})\left(\frac{\mathfrak{C}}{\mathfrak{B}^2}\times\mathfrak{C}\right) = HW\mathfrak{C}^2. \tag{20}$$

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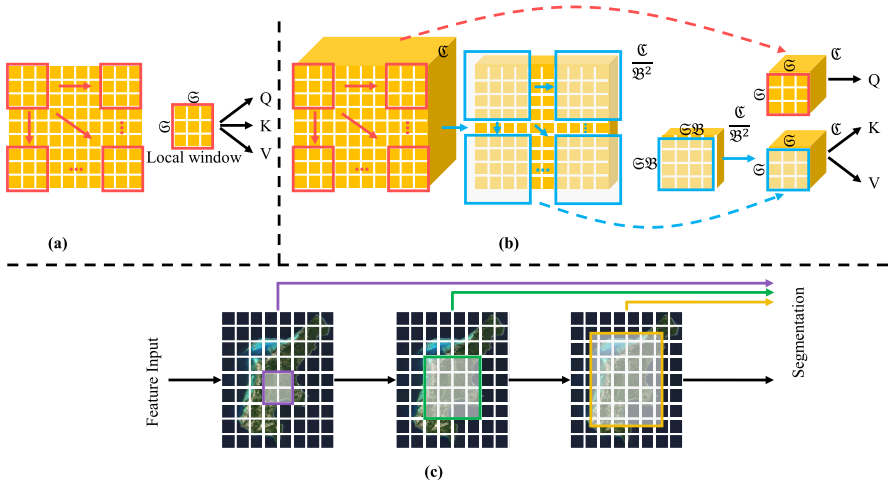


Fig. 6. Implementation of attention mechanisms in CR Transformer. (a) Local window attention. (b) Big Box attention. Q is derived from the local window transformation, while K and V are rescaled based on the big box. (c) Proposed SBBSPP network.

In terms of quantitative analysis, given the input feature $\mathbb{R}^{H\times W\times \mathfrak{C}}$, the computational cost of the Big Box attention is determined by

$$\begin{aligned}&\text{Cost(Big Box attention)}\\&=\left(2+\mathfrak{B}^2\right)(HW)\mathfrak{C}^2\\&\quad+2(HW)(\mathfrak{S}\mathfrak{B})^2\mathfrak{C}\approx 5(HW)\mathfrak{C}^2+2(HW)\mathfrak{S}^2\mathfrak{C}\end{aligned}\tag{21}$$

where the computational cost can be divided into two terms: the first term involves linear mappings [i.e., query (Q), key (K), value (V), and output operations ($\mathcal{L}$ and $\mathcal{S}$)], while the second term relates to attention computation (i.e., calculating attention matrices and weighted summation of the value).

E. CR Transformer Semantic Segmentation Model for Coral Reef Areas Benthic Mapping

The comprehensive structure of the CR Transformer semantic model is illustrated in Fig. 7. Given the assessment of hierarchical vision transformer (ViT) benefits, MiT [46] and Swin-Transformer [47] have been chosen to function as the encoder for the CR Transformer. MiT, an innovative blend of simple efficiency and empowering ViT [27] dedicated to semantic segmentation, is set to act as SegFormer's encoder. On the other hand, the Swin-Transformer is an exceptionally triumphant hierarchical ViT, rooted in local window attention. Specifically, the CR Transformer primarily processes two types of input data: original data represented by multichannel images (RGB, Nir, bathymetry derived from the Bathymetry Transformer, $b_b(\lambda)$, $a(\lambda)$, and benthic reflectance) and band calculation data. Table S1 (refer to Supplementary material S2) provides comprehensive information regarding the input data for the CR Transformer.

Before implementing SBBSPP, we concatenate the multilevel features of various output strides (i.e., the last three level features in the encoder part). These are resized to align with the feature dimension with an output stride of 8, and then, a linear layer is utilized to transform the concatenated features. Subsequently, the transformed feature, a byproduct of the previous process, is channeled into the SBBSPP, which yields the feature enriched with multiscale contextual information.

Within the most advanced ViTs used for semantic segmentation, the feature instrumental for the ultimate prediction of segmentation logits is unceasingly derived from the four-level features within encoders. Thus, we harness the first-level feature with an output stride of 4 to counteract the deficiency of low-level information. The resultant output from SBBSPP is enlarged to a size that matches a quarter of the input image. Following this, it combined with the first-level feature via a linear layer. Ultimately, the segmentation logits are predicted based on the feature that has been fortified at a low level.

Experiments on both the CR Transformer and SOTA models were pushed to their maximum accuracy under identical circumstances, following the same training strategy using the NVIDIA Telsa P100 cluster.

F. Evaluation Metrics

To gauge the effectiveness of remote sensing semantic segmentation for coral reefs at different water depths, the standard evaluation metrics are defined as follows:

$$\text{RMSE}=\sqrt{\frac{\sum_{i=0}^{N}\left(x_i^{\text{observed}}-x_i^{\text{predicted}}\right)^2}{N+1}}\tag{22}$$

$$\text{Accuracy}=\frac{\text{TP}_c+\text{TN}_c}{\text{TP}_c+\text{TN}_c+\text{FP}_c+\text{FN}_c}\tag{23}$$

$$\text{Precious}=\frac{\text{TP}_c}{\text{TP}_c+\text{FP}_c}\tag{24}$$

$$\text{Recall}=\frac{\text{TP}_c}{\text{TP}_c+\text{FN}_c}\tag{25}$$

$$\text{Dice}=\frac{2\text{TP}_c}{2\text{TP}_c+\text{FP}_c+\text{FN}_c}\tag{26}$$

$$\text{IoU}=\frac{S_{\text{Pred}}\cap S_{\text{Ground}}}{S_{\text{Pred}}\cup S_{\text{Ground}}}\tag{27}$$

$$\text{mIoU}=\frac{1}{N+1}\sum_{n=0}^{N}\text{IoU}\tag{28}$$

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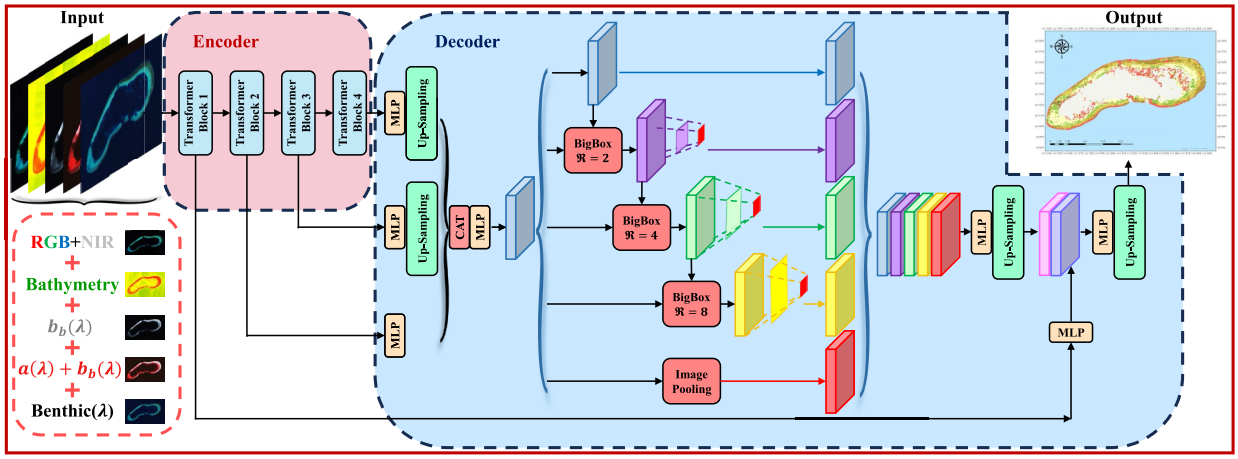


Fig. 7. Structure of CR Transformer semantic segmentation model.

where x_i^{observed} and $x_i^{\text{predicted}}$ $(i = 0, 1, 2, \ldots, N)$ represent the observed values (or actual values) and the predicted values, respectively; TP_c denotes true positives; TN_c denotes true negatives; FP_c denotes false positives; and FN_c denotes false negatives. Intersection over union (IoU) assesses the overlap between the predicted segmentation area S_{Pred} and the ground-truth segmentation area S_{Ground}, while mean IoU (mIoU) calculates the average IoU across multiple classes $(n = 0, 1, 2, \ldots, N)$.

IV. Results and Discussion

This section showcases the proposed method's derived results of water depth in coral reef regions and the semantic partitioning of coral reef substrates. It also analyzes the segmentation results of coral reefs at different depths, corroborating the relationship of water depth and coral reef segmentation consistency.

A. Water Depth Retrieval Results in Coral Reef Marine Areas

Fig. 8 displays the high-precision bathymetry results for different coral reef sea areas, obtained by the Bathymetry Transformer model. Fig. 8(a) presents the detailed bathymetric results of Rose Atoll, simultaneously also signifying its geomorphic features. Most of the areas in Rose Atoll bear a depth ranging between 0 and 4 m, exhibiting pronounced reef ridge characteristics. The atoll's landscape expands with a reef slope from the central point outward. At the fringes of the atoll, the water depth can extend up to more than 12 m. Even with the complex geomorphology and the abundance of islands and reefs in the Xisha Islands, our Bathymetry Transformer can generate highly precise depth results. The bathymetric span for the Yongle Islands, Huaguang Reef, and Panshi Island (shown in Fig. 8(b)–(d), respectively) encompasses 0–12 m, accurately presenting even the dense reefs submerged more than 10 m beneath the lagoon. As for Guam, distinguished by its streamlined coastlines that are portrayed in Fig. 8(e), it predominantly sees water depths of 0–3.5 m, reaching 12 m in the vicinity of the coral reef community at the southwest corner.

Notably, irrespective of being targeted at the high spatial resolution of PlanetScope or the moderate spatial resolution of Sentinel-2, the Bathymetry Transformer model always produces outcomes with a refined spatial resolution.

These results further indicate that whether it is open oceans or marginal seas, our Bathymetry Transformer model is capable of inversely deriving precise and low-error depth results, providing a new solution for obtaining water depths at large-scale, hard-to-reach locations.

B. Effect of the Bathymetry Transformer Model

To distinguish the influence of alterable water depths most effectively on the Bathymetry Transformer's preciseness, we juxtaposed the extracted bathymetry with the sampled depths at identical spots for validation. In Fig. 9, the multicolored points represent sampled data from extensive research zones, and the yellow squares denote verification sections for Bathymetry Transformer model outcomes. Furthermore, the marginal distribution plots of the comparative results are exhibited.

It is evident that the water depths obtained from the Bathymetry Transformer model are essentially distributed near the 1:1 line with the water depths from the verification data. To be specific, the Bathymetry Transformer model has achieved significant results at diverse locations (varying water depths). For the 0–2-m depth range, the RMSE is 0.117 m (Fig. 9(d)—Panshi Island). In the 0–3-m depth range, the RMSE is 0.176 m (Fig. 9(a)—Rose Atoll and Fig. 9(e)—Guam). Within the 0–4-m depth range, the RMSE is 0.218 m (Fig. 9(c)—Huaguang Reef). Moreover, the 0–5-m depth range exhibits an RMSE of 0.212 m (Fig. 9(b)—Yongle Islands). Overall, in the water depth range of 0–12 m, the Bathymetry Transformer model has achieved high-precision depth results with an RMSE of 0.375 m.

In addition, the Bathymetry Transformer exhibits more skill and robustness in the inverse bathymetric modeling of shallow waters compared to deep waters. For instance, in the 15-month lead-time prediction-verification comparison, the RMSE of the calculated water depth results for the majority of the upper shallow waters (less than 5 m) in the research area always

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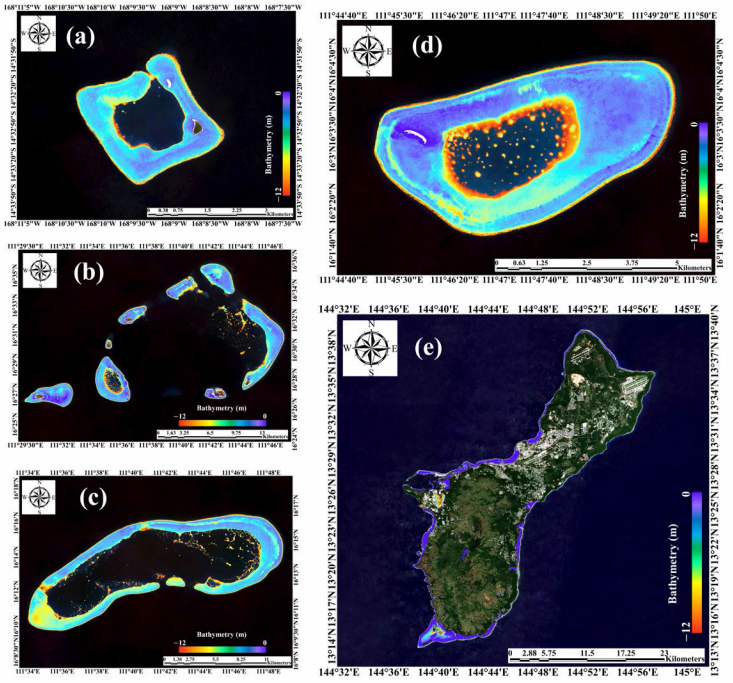


Fig. 8. Water bathymetric maps derived from the Bathymetry Transformer model. (a) Rose Atoll. (b) Yongle Islands, (c) Huaguang Reef, and (d) Panshi Island of Xisha Islands, in a north-to-south sequence. (e) Guam Island.

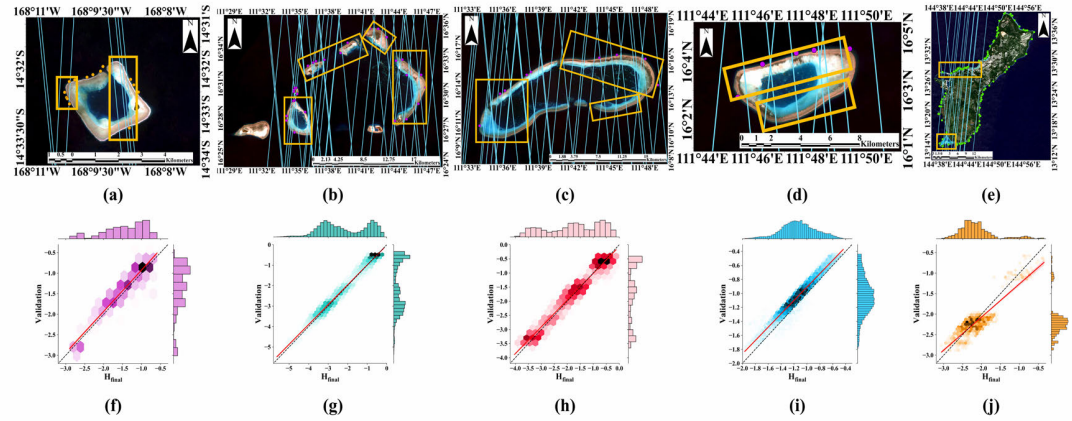


Fig. 9. Error scatter charts of the disparities between retrieved water depths (H_{final}) and validation depths across multiple coral reef marine regions. (a) Rose Atoll, (b) Yongle Islands, (c) Huaguang Reef, (d) Panshi Island of Xisha Islands, and (e) Guam Island with error charts (f)–(j), respectively.

remained less than 0.22 m, whereas in the deeper waters (5–12 m), the inverse depth results were less than 0.26 m, that is to say, our Bathymetry Transformer exhibits the capacity to provide corresponding high-precision depth measurements for optical remote sensing instruments, regardless of them being star-loaded satellites or airborne hyperspectral sensors. This competency is significantly beneficial in assisting depth analysis in the spectral characteristics of benthic habitats, especially for coral reef dwellers, which are extensively found within the depth range of 1–6 m.

C. Coral Reef Benthic Segmentation Results Across Varying Water Depths

Theoretically, water currents transport and deposit sand, resulting in a generally flat surface morphology with fewer attached vegetation and organisms, and minimal disturbances. This makes the spectral information of sand highly sensitive in capturing attribute variations. In contrast, identifying reefs, algae, and stony bubbles is more difficult due to their intricate coral reproduction and degradation processes, leading to mixed distributions.

From a practical standpoint, the CR Transformer, in conjunction with accurate preprocessed depth data and the inverted benthic reflectance in coral reef areas, effectively handles the issue of dense benthic mixing. As depicted in Fig. 10, the CR Transformer can finely segment different coral benthic types, aligning well with the field detection results at the sample sites. In addition, the CR Transformer combines the depth results from the Bathymetry Transformer to achieve precise differentiation between marine and benthic types, as exemplified in the coastal region.

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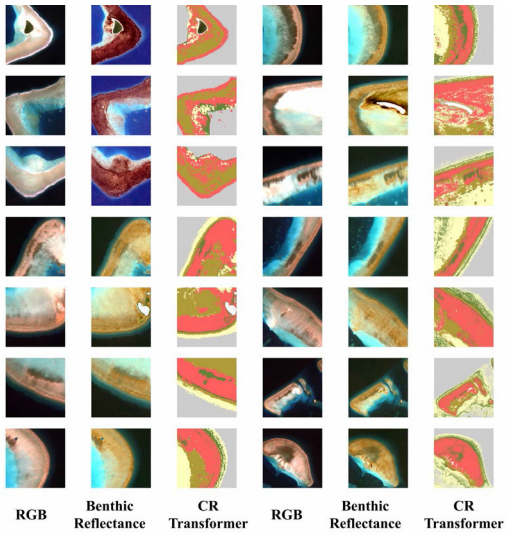


Fig. 10. Visualization of coral reef benthic segmentation results of the proposed method.

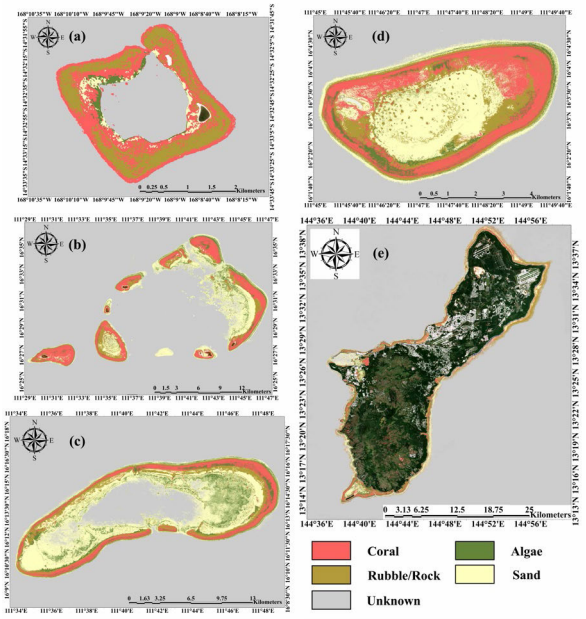


Fig. 11. Coral reef benthic mapping results of the CR Transformer. (a) Rose Atoll. (b) Yongle Islands, (c) Huaguang Reef, and (d) Panshi Island of Xisha Islands, in a north-to-south sequence. (e) Guam Island.

D. Coral Reef Benthic Mapping Results

Leveraging broad spanning, high-precision bathymetric readings, as well as benthic reflectance outcomes for coral reefs, the CR Transformer, backed by diverse input data analytics and optical remote sensing principles, handles each area's data to produce semantic partitioning for five distinct benthic categories. These segmentation outcomes, once assembled, manifest a complete depiction of the benthic mapping for coral reefs across the various study zones.

As depicted in Fig. 11, each study area is surrounded or enmeshed by the leading category, terming it Unknown and being mostly marine. Apart from the Unknown category, each research area displays a diverse range of coral reef benthic habitat environments. Fig. 11(a) exhibits the benthic division results for Rose Atoll where the large quantities of rock/rubble essentially structure the island's reef. Primarily, corals are distributed along the edge of the Atoll. The northwest inner region of Rose Atoll predominantly features sand and algae. Fig. 11(b)–(d), respectively, displays the three reefs of the Paracel Islands in China, namely, Yongle Islands, Huaguang Reef, and Panshi Island. The benthic type of these islands is primarily characterized by sand (apart from "Unknown"). The typical distribution follows a stratal structure from inside to outside, that is: sand–rubble/rock–coral–rubble/rock–sand. It is noteworthy that the Yongle Islands [Fig. 11(b)] consists of a series of small, dispersed reefs. Corals are extensively distributed across these islands, while rocks, commonly employed for wind resistance, are found predominantly on the eastern side and the outer ring of the reefs. Besides that, the isolated coral reefs in the circular lagoon [Fig. 11(d)], along with the underwater sandy grounds, fundamentally sculpt the complex benthic distribution layout of the coral reefs in the Xisha Islands of China. As for Guam [Fig. 11(e)], it is renowned for its elongated coral reef coastline, with a dense concentration of coral reefs particularly located at the southwest corner. The benthic genre in the nearshore region of Guam is largely dominated by rock/rubble, which is followed by coral, algae, and finally sand.

Regardless of whether the data are from PlanetScope or Sentinel-2, the CR Transformer consistently demonstrates exceptional and comprehensive mapping results, and the results are visually presented in Fig. 11. It is worth noting that the CR Transformer has the capability to go beyond the surface view depicted in remote sensing imagery. This is evident in Fig. 11(d), showcasing dense coral reefs underwater, and Fig. 11(b), illustrating underwater sand dunes.

E. Ablation Experiments of CR Transformer Semantic Segmentation Model

To evaluate the effectiveness of different components in boosting the performance of the CR Transformer, the ViT was used as a baseline model, and ablation experiments were conducted on the coral reef validation dataset.

Table II represents the superior performance of the CR Transformer model. Notably, the SBBSPP sets the record with an improvement in the model's overall accuracy of 4.85%. Following this, MIT block and SBBSPP remarkably reduce the model's GFLOPs 29-fold while managing to boost the mIoU accuracy by 3.79%, ultimately attaining an impressive 90.17% and exceeding the 90% mark.

Furthermore, as depicted in Table III, to evaluate the impact of various input data types on the final model output, we utilized the RGB + NIR category as a benchmark, incrementally adding input data types. Under benchmark conditions, the CR Transformer achieved a 90.10% mIoU. Under full-fledged conditions, where all data were used as input, our model reached 91.25% mIoU, marking a 1.15% improvement from the benchmark.

Significantly, both bathymetry and benthic reflectance play crucial roles in the escalation of data enhancement, with distinct slight increases in accuracy (increases of 1.61% and

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TABLE II
ABLATION EXPERIMENTS WITH VARIOUS NETWORK MODULES OF THE PROPOSED CR TRANSFORMER MODEL

Methods	#Parameters (M)	GFLOPs ↓	Accuracy (%) ↑	mIoU (%) ↑
Baseline: Vision Transformer (with MLP)	307	19365	90.11	86.38
+MiT Block	191.3	653	91.53	88.64
+SBBSPP	209.4	661	94.96	90.17
CR Transformer	211.7	670	**95.71**	**91.25**

TABLE III
ABLATION EXPERIMENTS WITH VARIOUS DATA SOURCES OF THE PROPOSED CR TRANSFORMER MODEL

Data sources	#Parameters (M)	GFLOPs ↓	Accuracy (%) ↑	mIoU (%) ↑
Benchmark: RGB+NIR	211.7	367	92.26	90.10
+Bathymetry: H	211.7	382	93.87	90.45
+$b_b(\lambda)$	211.7	386	93.88	90.45
+$a(\lambda)$	211.7	389	93.88	90.46
+Benthic(λ)	211.7	393	94.61	90.87
+Average	211.7	397	94.84	90.87
+Brightness	211.7	405	94.89	90.97
+NDWI	211.7	410	95.37	91.16
+D_b	211.7	459	95.59	91.21
+D_c	211.7	493	95.64	91.21
+$\oint s_*$	211.7	670	**95.71**	**91.25**

0.73%, respectively) and mIoU (increases of 0.35% and 0.41%, respectively).

F. Comparison Analysis of CR Transformer Semantic Segmentation Model With Existing Methods

The CR Transformer, along with 12 other SOTA models, has been implemented on the dataset. This collection of SOTA models encompasses both CNN-based [48], [49], [50], [51] and Transformer-based [45], [46], [47], [52], [53], [54], [55], [56] approaches. As shown in Fig. 12, our model occupies a prominent spot at the top center. It is demonstrated in Table IV that on the validation dataset, the proposed CR Transformer attains a superior position with an impressive mIoU of 91.25% and an accuracy of 95.71%.

Compared to the CNN-based model DNL [50], exhibiting top-tier accuracy, and OCRNet [51], the model with the fewest parameters, our CR Transformer steadfastly maintains its leadership position. It transcends DNL in the segmentation mIoU by 8.48% and accomplishes nearly equivalent light computation (as indicated by the GFLOPs) as OCRNet.

When pitted against Transformer-based models, our CR Transformer consistently sustains an upper hand, overtaking Swin-L UperNet [47] by an estimated 0.17% in mIoU. It also boasts a 6.3× swifter operation, indicative of a reduced latency. Moreover, our CR Transformer exhibits less GFLOP complexity, thus accelerating the segmentation speed to a level comparable to the fastest model, FastSAM [56].

Operating under the same experimental conditions, and contrasted with OneFormer [54] and Lawin Transformer [45], which have recently achieved impressive results, our model presents segmentations that are distinctly more outstanding. With near-identical parameter volumes, the accuracy indices of the CR Transformer surpass those of OneFormer and Lawin Transformer by 4.50% and 2.49%, respectively. Even in situations where a particular benthic species densely congregated with other types, CR Transformer consistently addresses these circumstances accurately, providing high-precision segmentation results in a considerably efficient manner.

Based on the evidence furnished in Figs. 11 and 12 and displayed in Table IV, the CR Transformer demonstrates excellent pixel-level segmentation precision. It effectively tackles the challenges of coral reef benthic mapping amidst varying water depths and different water column conditions.

G. Potential and Limitations of the Proposed Method

Our method combines both active and passive remote sensing data. Initially, we take advantage of a vast amount of PlanetScope, Sentinel-2, and ICESat-2 data to perform high-precision inversion for shallow water depth in coral reef areas, culminating in the production of the final high-quality depth results via the Bathymetry Transformer.

Subsequently, we fuse PlanetScope/Sentinel-2 images, water column IOPs features, benthic reflectance, and previously mentioned depth results, serving as input data for the CR Transformer to undertake semantic segmentation processes for coral reef benthic regions. A comprehensive amalgamation of different experiments has shown that our models consistently garner remarkable outcomes.

Both the Bathymetry Transformer and CR Transformer function as encoder–decoder models, fine-tuned for GPU training and inference, which skillfully make use of GPU resources to render highly precise outcomes. However, their performance might suffer restrictions when leveraged on edge devices.

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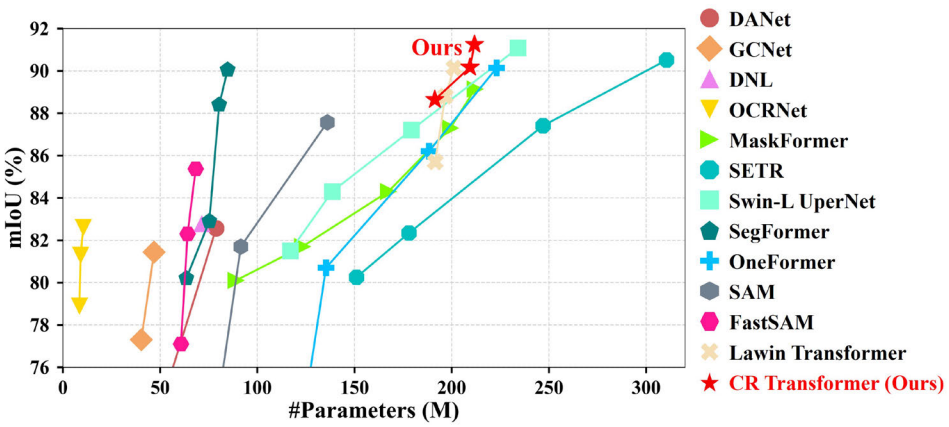


Fig. 12. Performances of CR Transformer and other SOTA semantic segmentation models.

TABLE IV
PERFORMANCES OF CR TRANSFORMER AND OTHER SOTA SEMANTIC SEGMENTATION MODELS

Model	Year@Publication	#Parameters (M)	GFLOPs ↓	Accuracy (%) ↑	mIoU (%) ↑
DANet	2019@CVPR	78.8	644	86.04	82.56
GCNet	2019@ICCV	46.9	641	85.28	81.44
DNL	2020@ECCV	71.5	1246	88.32	82.77
OCRNet	2020@ECCV	10.5	676	87.53	82.61
MaskFormer	2021@NeurIPS	212.0	712	91.17	89.14
SETR	2021@CVPR	310.5	1235	93.34	90.51
Swin-L UperNet	2021@ICCV	234.1	4263	94.29	91.08
SegFormer	2021@NeurIPS	84.7	439	92.27	90.05
OneFormer	2023@CVPR	223.1	2874	91.21	90.14
SAM	2023@Arxiv	136.0	1165	91.84	87.57
FastSAM	2023@Arxiv	68.3	342	90.11	85.37
Lawin Transformer	2023@CVPR	201.2	665	93.22	90.14
CR Transformer	2024	211.7	670	**95.71**	**91.25**

To boost their efficiency on gadgets operating with scarce GPU resources, our proposal is to trim the encoder of each Transformer or evaluate adopting a less resource-intensive structure in place of the primary encoders.

V. CONCLUSION

In this study, we present a novel method that pulls out information on water depth and benthic reflectance from coral reef areas, with the purpose of generating exceptionally accurate segmentation and mapping results of coral reef substrates. This method mainly consists of two novel Visual Transformers: Bathymetry Transformer and CR Transformer. In pursuing a deeper comprehension of the benthic distribution in coral reefs, we have consolidated a gamut of active and passive data streams, spanning ICESat-2 ATLAS, PlanetScope, Sentinel-2, and sample data into the Bathymetry Transformer for iterative training and validation procedures, eventually yielding highly precise water depth determinations. Following this, precious bathymetry from Bathymetry Transformer, supplemented by remote sensing band computation results, is injected into the proposed CR Transformer model. By combining the outcomes of prior processing stages, the CR Transformer employs MiT for hierarchical data processing to extract features at various scales. Subsequently, the novel SBBSPP concatenates multilevel features and conducts further fusion of contextual feature information. Ultimately, leveraging the integration of high-level semantic features and low-level semantic features, the CR Transformer produces benthic segmentation and mapping results.

A multitude of experiments bears testimony to the fact that our Bathymetry Transformer and CR Transformer models have, respectively, demonstrated remarkable achievements in the field of high-precision bathymetry retrieval and benthic segmentation accuracy. Each of these models exemplifies tangible structural efficiency, securing a leading position among the assortment of SOTA models. Remarkably, our Bathymetry Transformer can produce high-precision, high spatial-resolution bathymetry outcomes. It sets a record low of 0.375-m RMSE for the coral reef sea scope that spans from the surface to the 12-m water depth. With its compact and efficient design, the CR Transformer showcases an impressive accuracy of 95.71% and an mIoU of 91.25% on the coral reef validation dataset. In addition, it maintains superior segmentation precision compared to other SOTA models. Furthermore, the results gleaned from the validation datasets lend further support to the potency of our proposed method in segmenting benthic types across various scales and densely clustering ones. Broadly speaking, our Bathymetry Transformer and CR Transformer models steadfastly maintain front-runner positions in comparison to recent SOTA strategies.

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In summary, our findings underline the importance of measuring and considering both bathymetry and benthic mapping procedures during information retrieval in coral reef ecosystems. In our future endeavors, we will assimilate numerous academic innovations, relentlessly optimize the Bathymetry Transformer and CR Transformer, and drive the cooperative development between our models and their real-world applications.

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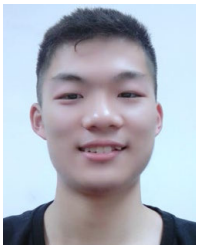
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